

NAUKRI JOB MARKET ANALYSIS DASHBOARD

Power BI Dashboard Project | Job Market & Salary Analytics

Role: Data Analyst (Independent Project)

Tools Used: Power BI, DAX, Power Query (M), Geo-mapping (Bing Maps)

Dataset: Naukri.com job postings dataset — 1,000 job listings across 6 departments, 8 companies, and 8 cities

Deliverable: Interactive multi-page Power BI dashboard with department/company slicers, salary benchmarking, trend analysis, and geo-visualization

1. Project Overview

This project analyzes job market data sourced from Naukri.com, India's leading job portal, to uncover hiring trends, salary benchmarks, and work-mode distribution across departments, companies, and cities. An interactive Power BI dashboard was built to help stakeholders — recruiters, job seekers, or HR analysts — quickly identify where hiring activity and compensation are concentrated, and how postings have trended over time.

2. Objective

- **Consolidate** raw job posting data into a single interactive view covering department, company, city, and time dimensions.
- **Benchmark** average salary (LPA) across departments and cities to surface pay disparities.
- **Analyze** work-mode distribution (Remote, Hybrid, On-site) across the job market.
- **Track** posting volume trends by month to identify hiring seasonality or slowdown.
- **Visualize** job distribution and salary geographically by state/city using a map.
- **Enable** self-service filtering by Department and Company for targeted exploration.

3. Dataset & Scope

Metric	Value
Total Jobs	1,000
Average Salary	13.40 LPA
Departments Covered	AI/ML, Data Analytics, Finance, HR, Marketing, Software (6)
Companies Covered	Accenture, Amazon, Capgemini, Deloitte, Flipkart, Infosys, TCS, Wipro (8)
Cities Covered	Gurugram, Pune, Chennai, Hyderabad, Kolkata, Bengaluru, Mumbai, Delhi (8)
Work Modes	Remote, Hybrid, On-site
Time Range Tracked	12 months (Jan–Dec, posting volume by month)

4. Dashboard Design

The dashboard was structured into three functional zones for guided, drill-down analysis:

- **Filter Panel (left rail):** global slicers for Department (AI/ML, Data Analytics, Finance, HR, Marketing, Software) and Company (Accenture, Amazon, Capgemini, Deloitte, Flipkart, Infosys, TCS, Wipro), driving every visual on the page.
- **KPI & Distribution Band:** a work-mode pie chart (Remote/Hybrid/On-site split), an Average Salary KPI card (13.40 LPA), and a Total Jobs KPI card (1,000).
- **Analytical Core:** bar charts for average salary by department and by city, a treemap of total jobs by department, a line chart tracking job postings by month, and a geo-map layering total jobs, average salary, and city location by state.

5. Key Insights

5.1 Work Mode Distribution

- **Work modes are near-evenly split:** Hybrid leads narrowly at 336 postings (33.6%), followed by On-site at 334 (33.4%) and Remote at 330 (33%) — indicating no single work arrangement dominates the current job market for these roles.

5.2 Salary Benchmarking by Department

- **Finance offers the highest average salary** at 14.5 LPA, followed closely by Data Analytics (13.6 LPA) and HR (13.4 LPA).
- **Marketing trails at 12.5 LPA**, the lowest among the six departments — a gap of 2 LPA versus Finance, useful for benchmarking compensation strategy or negotiating offers.
- **AI/ML and Software** sit in the middle of the range (13.2 LPA each), despite AI/ML typically commanding a premium in the broader market — worth a deeper look at role seniority mix within this dataset.

5.3 Salary Benchmarking by City

- **Gurugram (14.1 LPA) and Pune (14.0 LPA)** lead all cities in average salary, slightly ahead of Chennai and Hyderabad (13.7 LPA each).
- **Delhi has the lowest average salary** among the eight cities at 12.3 LPA, despite being a major metro — suggesting either a different job/department mix or more entry-level postings concentrated there.

5.4 Hiring Volume by Department

- **Software leads total job postings (186)**, followed closely by HR (176), Data Analytics (167), and AI/ML (169) — these four departments account for the bulk of hiring activity.
- **Finance (151) and Marketing (151)** have the fewest postings, tying for last — interesting to contrast against Finance's top salary ranking, implying fewer but higher-paying Finance roles.

5.5 Posting Trend Over Time

- **A clear downward trend** in job postings is visible from July (99 postings) through August (71 postings), the lowest point in the tracked period — a roughly 28% decline, suggesting a cooling hiring cycle or seasonal slowdown worth investigating against external market conditions.

5.6 Geographic Distribution

- **Hiring activity is concentrated in North and West India** (Delhi-NCR, Gurugram, Pune, Mumbai) alongside South Indian tech hubs (Bengaluru, Chennai, Hyderabad), consistent with India's established IT/corporate corridors.

6. Business Recommendations

- Use Finance and Data Analytics as benchmark departments when structuring competitive salary bands, given their consistent lead in average LPA.
- Investigate the July–August posting decline to determine whether it reflects a seasonal hiring lull or a genuine market slowdown, and plan recruitment campaigns accordingly.
- Re-examine AI/ML compensation positioning — its mid-pack average salary is lower than typically expected for the function and may signal a role-seniority skew in the dataset.
- Prioritize Gurugram and Pune for high-value role placements given their leading salary benchmarks.
- Leverage the near-even Remote/Hybrid/On-site split to offer flexible work-mode options as a differentiator in job postings, since no single mode dominates candidate supply.

7. Skills Demonstrated

- Dashboard design & data storytelling in Power BI
- DAX measures for KPIs (Total Jobs, Average Salary)
- Data modeling across department, company, city, and time dimensions
- Geo-spatial visualization (state/city-level mapping with Bing Maps integration)
- Trend analysis and time-series interpretation (monthly posting volume)
- Business insight generation and actionable recommendation writing